

Nigel L. E. Hew

Department of Materials Science and Engineering, The Pennsylvania State University
Steidle Building, University Park, PA 16802

☎ 814-769-6604 | ✉ njh5724@psu.edu | [in linkedin.com/in/nigelhew](https://www.linkedin.com/in/nigelhew) | github.com/nhew1994

Education

The University of Western Australia – Microelectronics Research Group 2022

Doctor of Philosophy

- Thesis: Atomistic Simulations of Dislocations in $\text{Hg}_x\text{Cd}_{1-x}\text{Te}$ and Related Compounds
- Advisors: Dr. Lorenzo Faraone and Dr. Dino Spagnoli
- Examiners: Dr. Nigel Marks and Dr. Christoph Grein

The University of Western Australia 2018

Master of Professional Engineering – Chemical Engineering

- Graduated with Distinction

The University of Western Australia 2016

Bachelor of Science – Engineering Science

Research & Work Experience

The Pennsylvania State University - Phases Research Lab June 2023 - Present

Postdoctoral Scholar

- Performed density functional theory (DFT) calculations within the quasi-harmonic approximation to model finite-temperature thermodynamic properties of magnetic and ferroelectric materials
- Applied recursive entropy to first-principles thermodynamics to predict emergent phenomena such as negative thermal expansion
- Developed open-source Python tools on GitHub, including Density Functional Theory ToolKit (DFTTK) to automate quasi-harmonic approximation workflows with VASP, and pyentropy to implement recursive entropy in first-principles thermodynamics
- Generated DFT energy-volume curve datasets to train and fine-tune a machine learning model for FCC high-entropy alloys
- Performed computational fluid dynamics (CFD) simulations to model humping behavior in laser welding of stainless steel plates
- Performed molecular dynamics (MD) simulations to calculate viscosity and short-range ordering in $\text{Al}_{1-x}\text{Cu}_x$ alloys
- Drafted major sections of DOE, NSF, and ACS grant proposals

Research Associate

- Generated DFT transition-level diagrams for point defects in $\text{Hg}_{1-x}\text{Cd}_x\text{Te}$
- Set up deep-level transient spectroscopy (DLTS) experiments to measure point defect levels in $\text{Hg}_{1-x}\text{Cd}_x\text{Te}$

Publications

Journal Publications

1. Lai, Z.-H., **Hew, N. L. E.**, Xu, S. & Li, J. Molten pool dynamics and humping suppression in high-speed laser welding via tailored beam configurations. *submitted*.
2. Linton, N., **Hew, N. L. E.**, Liu, Z.-K. & Aidhy, D. S. Machine Learning Framework for Predicting EnergyVolume Curves in FCC High-Entropy Alloys. *in preparation*.
3. **Hew, N. L. E.**, Myers, L. A., Shang, S.-L. & Liu, Z.-K. pyzentropy: A Python package implementing recursive entropy for first-principles thermodynamics. *arXiv:2604.17665 [cond-mat.mtrl-sci]* (2026).
4. **Hew, N. L. E.**, Myers, L. A., van de Walle, A., Shang, S.-L. & Liu, Z.-K. Density Functional Theory ToolKit (DFTTK) to automate first-principles thermodynamics via the quasiharmonic approximation. *Computational Materials Science* 258, 114072 (2025). [10.1016/j.commatsci.2025.114072](https://doi.org/10.1016/j.commatsci.2025.114072)
5. Myers, L. A., **Hew, N. L. E.**, Shang, S.-L. & Liu, Z.-K. Recursive entropy in thermodynamics: establishing the statistical-physics basis of the zentropy approach. *arXiv:2511.04950 [cond-mat.stat-mech]* (2025).
6. Myers, L. A., **Hew, N. L. E.**, Shang, S.-L. & Liu, Z.-K. Stripe antiferromagnetic ground-state configuration of FeSe revealed by density functional theory. *Physical Review B* 111, L121109– (2025). [10.1103/PhysRevB.111.L121109](https://doi.org/10.1103/PhysRevB.111.L121109)
7. Shang, S.-L., **Hew, N. L.**, Gong, R., Cockrell, C., Bingham, P. A., Guo, X., Li, J., Hong, Q.-J. & Liu, Z.-K. Achieving accurate entropy and melting point by ab initio molecular dynamics and zentropy theory: Application to fluoride and chloride molten salts. *Journal of Molecular Liquids* 438, 128651 (2025). [10.1016/j.molliq.2025.128651](https://doi.org/10.1016/j.molliq.2025.128651)
8. **Hew, N. L. E.**, Shang, S.-L. & Liu, Z.-K. Predicting phase transitions in PbTiO_3 using zentropy through quasiharmonic phonon calculations. *Physical Review B* 110, 184103 (2024). [10.1103/PhysRevB.110.184103](https://doi.org/10.1103/PhysRevB.110.184103)
9. Liu, Z.-K., **Hew, N. L. E.** & Shang, S.-L. Zentropy theory for accurate prediction of free energy, volume, and thermal expansion without fitting parameters. *Microstructures* 4 (2024). [10.20517/microstructures.2023.56](https://doi.org/10.20517/microstructures.2023.56)
10. **Hew, N. L. E.**, Spagnoli, D. & Faraone, L. First-Principles Study of the 30° and 90° Partial Dislocations in HgTe , CdTe , and $\text{Hg}_{0.7}\text{Cd}_{0.3}\text{Te}$. *Phys. Status Solidi B* 259, 2200180 (2022).

[10.1002/pssb.202200180](https://doi.org/10.1002/pssb.202200180)

11. **Hew, N.**, Spagnoli, D. & Faraone, L. Dislocation core energies of the 0° perfect, 60° perfect, 30° partial, and 90° partial dislocations in CdTe, HgTe, and ZnTe: A molecular statics and elasticity theory analysis. *Materials Today Communications* 26, 101949 (2021). <https://doi.org/10.1016/j.mtcomm.2020.101949>
12. **Hew, N.**, Spagnoli, D. & Faraone, L. Molecular Dynamics Study of Heteroepitaxial Growth of HgCdTe on Perfect and Dislocated (211)B CdZnTe Substrates. *ACS Applied Electronic Materials* 3, 5102–5113 (2021). [10.1021/acsaelm.1c00835](https://doi.org/10.1021/acsaelm.1c00835)

Presentations

Conference Presentations

1. Zentropy: Implementing Recursive Entropy for First-Principles Thermodynamics at the 2026 TMS Annual Meeting & Exhibition, San Diego.
2. PyZentropy: A Python package implementing recursive entropy for first-principles thermodynamics at the 2025 SIAM New York-New Jersey-Pennsylvania Section Conference, Pennsylvania State University, State College, PA.
3. DFTTK and PyZentropy: High-Throughput Tools for Free Energy Calculations of Individual Configurations and Systems from First-Principles at the 2025 TMS Annual Meeting & Exhibition, Las Vegas.
4. Using Zentropy to Predict Phase Transitions in PbTiO₃ at the 2024 TMS Annual Meeting & Exhibition, Orlando.
5. First-principles study of the 30° and 90° partial dislocations in HgTe, CdTe, and Hg_{0.7}Cd_{0.3}Te at the 2021 U.S. Workshop on the Physics and Chemistry of II-VI Materials, Chicago.
6. Molecular Simulations of Dislocation Energies in CdTe, HgTe and ZnTe at the 2019 U.S. Workshop on the Physics and Chemistry of II-VI Materials, Chicago, and at the 2019 20th International Union of Materials Research Societies International Conference in Asia, Perth.

Conference Posters

1. Molecular Dynamics Studies of Dislocations in CdTe, HgTe, and ZnTe at the 2019 RACI Australian Conference on Physical Chemistry, Perth.

Scholarships & Awards

The University of Western Australia

- Ad Hoc Postgraduate Scholarship and Research Training Program International Fees Offset from the Microelectronics Research Group at the University of Western Australia (2021-2022).
- Australian Government Research Training Program (2018-2021).

- University of Western Australia Safety-Net Top-Up Scholarship (2018).
- Chevron Chair Prize in Gas Process Engineering Unit Operations and Unit Processes (2015) awarded to the top five students in ENSC3019: Unit Operations and Unit Processes.

The Pennsylvania State University

- Penn State Office of Postdoctoral Affairs Travel Award (\$500) for presenting research at the TMS 2026 Annual Meeting, 2026.

Teaching & Mentorship

The Pennsylvania State University

Fall 2023

Teaching Assistant

- MATSE580: Computational Thermodynamics - Dr. Zi-Kui Liu
 - Designed and delivered four 75-minute lectures on density functional theory (DFT), VASP usage, and phonon calculations
 - Created and graded two homework assignments focused on DFT simulations using VASP, including phonon calculations

Student Mentorship

- Mentored multiple Ph.D. students in the Phases Research Lab, under Dr. Zi-Kui Liu

Service & Leadership

External

- Peer Reviewer, Physical Review B 2024 – current
- Peer Reviewer, Physical Review Letters 2026 – current

The Pennsylvania State University

- Rising Researcher Collaborations - ICDS 2025 – current
- Poster Judge - ICDS Symposium Oct 2025
- Poster Judge - Materials Day, MRI Oct 2025

Technical Skills

- Computational Tools: VASP, LAMMPS, FLOW-3D, SLURM
- Programming Languages: Python, MATLAB
- Version Control & Collaboration: GitHub
- Development Environments: VS Code
- Productivity Tools: Microsoft Office, Overleaf

Memberships

- The Minerals, Metals & Materials Society (TMS)

References

Dino Spagnoli, Ph.D.

Associate Professor, School of Molecular Sciences
The University of Western Australia
Perth, WA 6009, Australia

✉ dino.spagnoli@uwa.edu.au

Christoph Grein, Ph.D.

Professor, Physics
University of Illinois Chicago
Chicago, IL 60607, United States

✉ grein@uic.edu

Zi-Kui Liu, Ph.D.

Professor of Materials Science and Engineering
Department of Materials Science and Engineering
The Pennsylvania State University
University Park, PA 16802, United States

✉ liu@matse.psu.edu

Shun-Li Shang, Ph.D.

Research Professor
Department of Materials Science and Engineering
The Pennsylvania State University
University Park, PA 16802, United States

✉ sus26@psu.edu